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CLINICAL INVESTIGATION **Head and Neck**

RESULTS OF PROTON THERAPY OF UVEAL MELANOMAS TREATED IN NICE

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Purpose: To present the first results of uveal melanomas treated with the Medicyc Cyclotron 65 MeV proton beam facility in Nice, analyzing the factors that affect the cause-specific survival (CSS), metastatic rate, and reporting the visual outcome.
Methods and Materials: This study concerns 538 patients referred by French institutions between June 1991 and December 1996. The eye and tumor parameters were measured using ultrasonography and angiography. Since 1994, CT scans were performed in most patients to help determine the axial length and the shape of the ocular globe. Tantalum clips were inserted around the tumor by the referring ophthalmologist. There were 349 posterior pole tumors (64.9%), 130 equatorial tumors (24.1%), and 59 ciliary body tumors (11%). Two hundred four patients (37.9%) had T1 or T2 tumors, and 334 patients (62.1%) had T3 or T4 tumors. The median tumor diameter was 14.6 mm, and the median tumor height was 5.1 mm. All patients received 52 Gy (57.20 Gy Co-equivalent dose) on 4 consecutive days. The data were analyzed by December 1997.
Results: The CSS was 77.4% at 78 months, the overall survival was 73.8% and the local control was 89.0%. The CSS was not influenced by the patient age or the site of the tumor. It was 81.5% for T1 and T2 tumors, versus 75% for T3 and T4 tumors ($P = 0.035$). It was found that the tumor diameter, rather than the height, was the most important parameter affecting outcome. The metastatic rate was 8%. It depended on the T stage, tumor diameter and thickness, but not the tumor site. Thirty-eight enucleations were performed, most of them due to tumor progression and/or glaucoma. One-third of the patients in whom visual acuity was adequately scored before and after treatment had a stable, if not improved vision, and half the patients retained useful vision after treatment.
Conclusion: The outcome of patients suffering from uveal melanoma and treated with high-energy protons compares favorably with other techniques of treatment. The tumor dimensions affected CSS and metastatic rate. Even though two-thirds of patients had posterior pole tumors, half of them retained useful vision.

Proton therapy, Uveal melanoma, Eye tumors, Treatment, Cause-specific survival, Prognostic factors, Metastatic rate, Local control, Enucleation rate, Complications, Visual acuity.

INTRODUCTION

Conservative treatment of uveal melanoma includes brachytherapy with different radioactive sources, transscleral local resection, transpupillary thermotherapy, laser photocoagulation, and external particle radiotherapy (1). Proton-beam therapy of eye tumors offers the ideal physical properties to deliver a precise and homogeneous dose distribution within the target volume, with preservation of sensitive normal tissues due to the distal fall-off of the dose beyond the spread-out Bragg peak.

In the 1980s, a medical cyclotron (Medicyc) accelerating protons to 65 MeV was designed and installed in the Cancer Institute in Nice. Treatment of ocular tumors began on June 1991, and since then there has been a regular accrual of about 170 patients per year, mainly suffering from uveal melanomas. In this study, we report results of proton therapy in the first 538 patients referred by the French cooperative group. This paper focuses on the factors affecting the cause-specific survival (CSS) and the metastatic rate. Information on local control, enucleation rate, and the visual outcome of these patients are also given. Observations on factors affecting local control and eye retention were briefly reported elsewhere (2), and will be detailed in a subsequent paper.

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